Bharatiya Antariksh Hackathon (BAH) 2026

P.S 10: **Infrared Image Colorization and Enhancement for Improved Object Interpretation**

Transforming Low-Resolution TIR Satellite Images into High-Resolution TIR and Colorized RGB Satellite Image

Team NovaPixelX4

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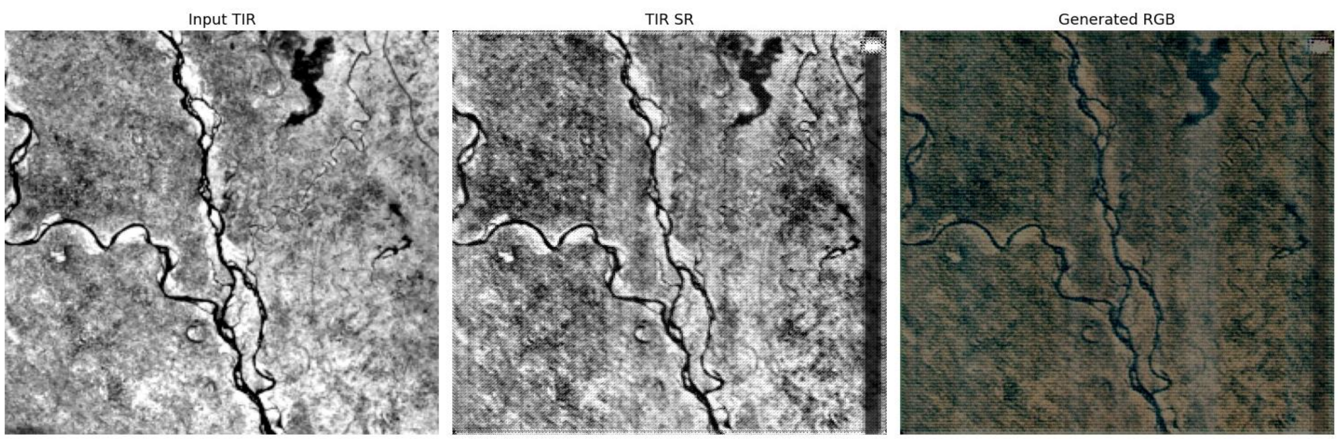
1. Executive Summary

Thermal Infrared (TIR) satellite imagery is a critical data source for environmental monitoring, disaster mitigation, wildfire detection, urban heat island analysis, volcanic monitoring, and night-time Earth observation. Spaceborne TIR sensors, however, are constrained by physical optics and sensor trade-offs, producing single-band grayscale imagery at markedly lower spatial resolution than the accompanying optical bands. This limits an analyst's ability to visually interpret and distinguish objects within the scene.

Team NovapixelX4 presents a dual-stage, physics-informed deep learning pipeline that first reconstructs a super-resolved thermal image from a low-resolution input, and then translates that enhanced thermal signal into a perceptually realistic, colorized RGB representation. Both stages are implemented as independent Pix2Pix Conditional Generative Adversarial Networks (cGANs), combining a U-Net generator with a PatchGAN discriminator.

The system was developed and evaluated using Landsat-9 imagery across three locations Delhi, Jammu, and Lucknow with two locations used for training and one (Jammu) held out for testing (Evaluation). On the 16-image Jammu test set, the pipeline achieves an **average PSNR of 18.56 dB** and an **average SSIM of 0.508**, figures discussed in detail in Section 11 along with their implications for output quality.



Result 1

2. Introduction and Motivation

Thermal Infrared sensors capture the radiant heat emitted by surface materials, providing information that is invisible to optical sensors fire fronts through smoke, subsurface moisture patterns, urban heat retention, and nocturnal terrain features. This information value is offset by two structural limitations of TIR data:

- Lower native spatial resolution than optical bands, driven by detector and optics constraints rather than orbital geometry.
- Single-channel grayscale output, which strips away the intuitive colour cues human analysts rely on for rapid object recognition.

This project addresses both limitations in sequence: a super-resolution stage recovers spatial detail lost to sensor constraints, and a colorization stage maps the enhanced thermal signal onto a synthetic RGB representation trained against real Landsat optical bands. The result is intended to make TIR scenes faster and more intuitive for a human analyst to interpret, without discarding the underlying radiometric information.

3. System Objectives

- Enhance Resolution — reconstruct sharp spatial structure from a 200 m thermal input to a 100 m output.
- Radiometric Consistency — preserve physically meaningful, calibrated temperature relationships in the generated output rather than treating pixels as arbitrary intensities.
- Realistic Translation — synthesize RGB colours that plausibly correspond to the physical thermal profile of surface materials.
- Analyst-Centric Design — produce output suited to rapid visual assessment in time-sensitive contexts such as disaster response.
- Preserve Structural Information — recover fine-grained boundaries and gradients typically lost to low-resolution sensing.

4. Dataset and Preprocessing

4.1 Source Data — Landsat-9

The pipeline is built on multispectral and thermal bands from the Landsat-9 mission. Three optical bands supply the RGB training target, and the thermal band supplies the model input:

Band	Spectral Domain	Role in Pipeline
B2	Blue (0.450–0.515 µm)	RGB target Channel 3
B3	Green (0.525–0.600 µm)	RGB target Channel 2
B4	Red (0.640–0.670 µm)	RGB target Channel 1
B10	Thermal Infrared (10.60–11.19 µm)	Primary physical input

Three geographic locations were used: Delhi, Jammu, and Lucknow. Two locations were used for training, with Jammu held out as an independent test location.

4.2 Input Directory Structure

```
input/
├── <Location_1>/
│   ├── <file_prefix_1>_B10.TIF
│   ├── <file_prefix_1>_B2.TIF
│   ├── <file_prefix_1>_B3.TIF
│   └── <file_prefix_1>_B4.TIF
├── <Location_2>/
│   └── ... (same 4 band files)
└── <Location_3>/
    └── ... (same 4 band files)
```

4.3 Preprocessing Pipeline

- Reading raw GeoTIFF satellite imagery for each band.
- Patch extraction from full scenes into fixed-size training samples.
- Synthetic downsampling of thermal patches to generate paired low-resolution / high-resolution training examples (200 m input, 100 m target).
- Percentile clipping, applied only for visualization purposes.
- Per-patch band normalization(Global normalization).
- Generation of paired NumPy (.npy) datasets, preserving original radiometric values rather than lossy image formats.

4.4 Generated Dataset Structure

```
output/
└── patches/
    ├── <Location_1>/
    │   ├── sample_000/
    │   │   ├── rgb_100m_512.npy
    │   │   ├── rgb_100m_512.png
    │   │   ├── tir_100m_512.npy
    │   │   ├── tir_100m_512.png
    │   │   ├── tir_200m.npy
    │   │   └── tir_200m.png
    │   ├── sample_001/
    │   │   └── ... (same 6 files)
    │   └── sample_015/
    │       └── ...
    └── <Location_2>/
        └── sample_000/
            └── ...
```

5. Methodology Overview

The complete system is composed of two independently trained deep learning stages, chained at inference time:

```
Low-Resolution Thermal Image (200 m) dim=(1,256,256)
            │
            ▼     pre- processing 1 Dimension change(2X)

Low-Resolution Thermal Image (200 m) dim=(1,512,512)
            │
            ▼     Model 1 (Super resolution)
High-Resolution Thermal Image (100 m)  — dim=(1,512,512) Stage 1: Super-Resolution
            │
            ▼     pre-processing 2  1 channel to 3 channel (extra physics feature)

High-Resolution Thermal Image (100 m)  — dim=(3,512,512) Stage 1: Super-Resolution
            │
            ▼     Model 2 (RGB convertor) Colorized TIR RGB Image (100 m)

 Colorized RGB Satellite Images
```

6. Stage 1 — Thermal Super-Resolution

6.1 Objective

Reconstruct a sharp, high-resolution (100 m) thermal image from a single-channel, low-resolution (200 m) input, recovering fine structural boundaries and gradients that are otherwise lost to sensor-level resolution limits.

6.2 Pipeline Flow

```
Input Low-Res TIR (200 m) — 1 × 256 × 256
              │
              ▼
Preprocessing: Bicubic Interpolation (scale-up ×2)
              │
              ▼
Spatially Upsampled TIR (blurry) — 1 × 512 × 512
              │
              ▼
Pix2Pix Generator (U-Net) — learns texture / detail map
              │
              ▼
Output High-Res TIR (100 m) — 1 × 512 × 512
```

6.3 Pre-processing of Thermal Image (For Model Training)

The output generated by the super-resolution network is subjected to a sequence of post-processing operations to improve its visual quality while preserving the radiometric characteristics of the original thermal infrared (TIR) image. These operations ensure that the generated image remains physically consistent and suitable for further analysis.

1. Min-Max Normalization (TIR dim(1 x 512 x 512))

The generated image is first normalized to the range ([0,1]) using Min-Max normalization:

$$I\_TIR = (I\_gen - min(I\_gen)) / (max(I\_gen) - min(I\_gen) + \varepsilon)$$

where ***I_gen*** denotes the generated image TIR(1x256x256) to **TIR(1x512x512)** and $\varepsilon = 10^{-6}$ is added to avoid division by zero. This step standardizes the pixel intensity range and improves numerical stability.

2. RGB to Single-Channel Conversion(will be used as a part of output)

Since the generator produces a three-channel RGB output while thermal images are inherently single-channel, the RGB image is converted into grayscale by averaging the three channels:

$$I\_gray = (R + G + B) / 3$$

This conversion produces a **single-channel thermal representation** suitable for subsequent processing.

3. Mean Intensity Alignment

The average brightness of the generated image is aligned with that of the original low-resolution thermal image to preserve its **radiometric characteristics (preserve thermal features)**.The adjustment is performed as:

$$I\_align = I\_gray - \mu\_gray + \mu\_TIR$$

where

- μ_gray is the mean intensity of the generated grayscale image.
- μ_TIR is the mean intensity of the original thermal image.

This operation ensures that the generated image maintains a brightness distribution similar to the original thermal data.

4. Weighted Image Fusion

To preserve the original thermal information while incorporating **enhanced spatial details**, the aligned image is blended with the original thermal image using weighted averaging:

$$I\_final = 0.3 \times I\_align + 0.7 \times I\_TIR$$

where *I_TIR* represents the original thermal image. A higher weight is assigned to the original image to retain authentic thermal signatures while increasing resolution features and reducing artifacts introduced by the generator.

5. Intensity Clipping

Finally, the pixel intensities are restricted to the valid range:

$$I\_output = clip(I\_final, 0, 1)$$

This step prevents underflow and overflow of intensity values, ensuring that the output image remains valid for visualization and further processing.

Summary

The proposed pre-processing pipeline consists of normalization, grayscale conversion, mean intensity alignment, weighted fusion, and intensity clipping. These operations collectively improve the perceptual quality of the generated high-resolution thermal image while preserving its original thermal characteristics. As a result, the final output exhibits enhanced spatial resolution with improved radiometric consistency, making it suitable for practical thermal image analysis and downstream computer vision applications. **Now original TIR low quality (1 x 512 x 512) and embedded (**waited fused ***I_output*** **) TIR(1 x 512 x 512) will trained by the model to get high resolution TIR.**

6.4 Model Architecture — Pix2Pix Conditional GAN

Rather than relying purely on mathematical interpolation, Stage 1 uses a conditional GAN that learns a direct mapping from the blurred, bicubically upscaled matrix to a crisply defined high-resolution target. The interpolated image serves as the conditioning input to both the generator and the discriminator.

- **Generator:** a U-Net with skip connections between mirrored encoder and decoder layers, propagating high-frequency spatial detail across the network bottleneck.
- **Discriminator:** a PatchGAN classifier, which scores realism at the level of local image patches rather than the image as a whole.

U-Net Generator — Structural Summary

Encoder Stage	Channels	Decoder Stage
E1: Conv	64	D1: ConvTranspose (64)
E2: Conv + BatchNorm	128	D2: ConvTranspose (128)
E3: Conv + BatchNorm	256	D3: ConvTranspose (256)
Bottleneck	512	Deep feature extraction core

Each encoder stage is linked to its mirrored decoder stage via a skip (identity) connection, preserving pixel-level spatial localization that would otherwise be lost through the bottleneck.

6.5 Advantages of This Approach

- Lightweight architecture relative to deeper SR networks.
- Fast inference suited to iterative experimentation.
- Efficient pixel-shuffle-style up-sampling.
- Learned reconstruction quality superior to fixed interpolation kernels.
- Practical to run without dedicated GPU infrastructure.

6.6 Model Checkpoints

- Weights directory: /checkpoints/
- Output naming schema: gen_SR_100.pth (generator), disc_SR_100.pth (discriminator)
- Separate checkpoint pairs are maintained for the Stage 1 (SR).

7. Stage 2 — Physics-Aware Thermal Colorization

7.1 Objective

Translate the Stage 1 output — enhanced 100 m thermal imagery, expressed as the 3-channel physics-informed tensor described in Section 4.5 — into a realistic RGB image, using a second, independently trained Pix2Pix conditional GAN.

7.2 Pipeline Flow

```
TIR 100m(1 x 512 × 512) Matrix
            │   Physics Feature Extraction
            ▼
3-Channel Input Tensor
  1.Normalized TIR (global min-max)
  2.Spatial rolling local mean (3×3 blur)
  3.High-pass spatial gradient magnitude
            │
            ▼
Pix2Pix Generator (U-Net, skip connections)
            │
            ▼
Generated Colorized TIR RGB Output
```

7.3 Physics-Informed Feature Extraction

Rather than passing the raw single-channel thermal array directly into the colorization network, the pipeline derives a three-channel physics-informed tensor from each thermal patch:

- **Channel 1:** Global Normalization: min-max scaling bounded by pre-calibrated limits (TIR_GLOBAL_MIN = 15000, TIR_GLOBAL_MAX = 35000) to maintain radiometric consistency across scenes.
- **Channel 2**: Thermal Diffusion Mapping: a rolling 3×3 local mean (implemented via np.roll along spatial axes) approximating localized heat dispersion.
- **Channel 3:** Spatial Thermal Gradient: an edge-detection field computed as $G = \sqrt{(g_x^2 + g_y^2)}$, highlighting structural boundaries and material interfaces.

This engineered 3-channel tensor rather than the raw single-band thermal image is what is fed to the Stage 2 generator, giving it explicit structural and radiometric cues alongside the intensity data.

7.4 Architecture

- Generator: U-Net architecture with skip connections and an encoder-decoder structure, structurally consistent with the Stage 1 generator.
- Discriminator: PatchGAN, evaluating 70×70 local patches as real or fake rather than scoring the full image at once.

7.5 Loss Formulation

The discriminator D is trained to distinguish genuine (physics-input, real-RGB) pairs from (physics-input, generated-RGB) pairs:

$$L_D = BCE(\ D(Physics,\ RGB),\ 1\ ) + BCE(\ D(Physics,\ Gen(Physics)),\ 0\ )$$

The generator G is trained on a composite, multi-objective loss combining adversarial, pixel-fidelity, and edge-coherence terms:

$$L^G = \alpha \cdot L\_GAN + \beta \cdot L1 + \gamma \cdot L\_edge$$

where α, β, and γ are scale weights balancing adversarial realism against pixel-level reconstruction accuracy and structural edge alignment.

7.6 Training Hyperparameters

Parameter	Value	Rationale
Batch size	1	Enforced by pixel-to-pixel structural dependency in paired image translation
Learning rate	2×10^{-4}	Standard Pix2Pix convergence schedule for Adam
Optimizer / Betas	Adam, (β_1=0.5, β_2=0.999)	Reduced β_1 momentum to reduce training instability
Epochs	100	Full sweep over the training patch set

7.7 Model Checkpoints

- Weights directory: /checkpoints/
- Output naming schema: gen_100.pth (generator), disc_100.pth (discriminator)
- Separate checkpoint pairs are maintained for Stage 2 (colorization) networks.

8. Complete End-to-End Pipeline

```
Input Thermal Infrared Image (200 m)
            │
            ▼
Thermal Image Normalization
            │
            ▼
Stage 1: Pix2Pix Super-Resolution
            │
            ▼
High-Resolution Thermal Image (100 m)
            │
            ▼
Physics Feature Extraction: (1, 512, 512) → (3, 512, 512)
            │
            ▼
Stage 2: Pix2Pix Colorization
            │
            ▼
Colorized RGB Image (100 m)
            │
            ▼
GeoTIFF / PNG / NPY Export
```

9. Implementation and Training

9.1 Installation

```
pip install torch torchvision numpy opencv-python pillow matplotlib tifffile tqdm scikit-image
```

Additional standard-library modules used (no installation required): os, glob, random, argparse, logging.

9.2 Execution Commands

Step	Command
Generate dataset	python models/driver.py
Train super-resolution model	python models/train_pix2pixSR.py
Train colorization model	python models/train_pix2pixCol.py
Evaluate trained model	python eval.py
Run inference on a test sample	python Run_model.py

9.3 Output Artifacts

```
results/
├── tir.npy / tir.png     (raw input thermal image)
├── SR.npy / SR.png       (super-resolved thermal image)
└── RGB.npy / RGB.png     (colorized RGB output)
```

10. Quantitative Evaluation and Results

Performance was evaluated on 16 held-out images from the Jammu test location, which was excluded from training.

Metric	Value	Interpretation
Average MSE	0.0356	Low pixel-wise error on a normalized [0,1] scale
Average MAE	0.1345	Moderate average absolute pixel deviation
Average PSNR	18.56 dB	Below the ~25–30 dB range typically associated with strong SR/colorization fidelity
Average SSIM	0.508	Indicates only moderate structural similarity to ground truth

```
✅ import done
sample_000            PSNR=21.24   SSIM=0.7725
sample_001            PSNR=25.41   SSIM=0.6614
sample_002            PSNR=24.71   SSIM=0.6927
sample_003            PSNR=27.03   SSIM=0.7646
sample_004            PSNR=20.80   SSIM=0.7144
sample_005            PSNR=10.92   SSIM=0.3344
sample_006            PSNR=10.51   SSIM=0.3063
sample_007            PSNR=11.66   SSIM=0.2339
sample_008            PSNR=24.67   SSIM=0.6493
sample_009            PSNR=10.45   SSIM=0.3028
sample_010            PSNR=8.82   SSIM=0.2866
sample_011            PSNR=19.72   SSIM=0.4750
sample_012            PSNR=22.40   SSIM=0.6152
sample_013            PSNR=28.21   SSIM=0.5783
sample_014            PSNR=13.42   SSIM=0.2308
sample_015            PSNR=17.06   SSIM=0.5136

========================
Evaluation Results
========================
Images : 16

MSE  -> Avg: 0.035639 | Min: 0.001509 | Max: 0.131353
MAE  -> Avg: 0.134521 | Min: 0.024684 | Max: 0.357780
PSNR -> Avg: 18.5644 | Min: 8.8156 | Max: 28.2124
SSIM -> Avg: 0.508233 | Min: 0.230778 | Max: 0.772518
```

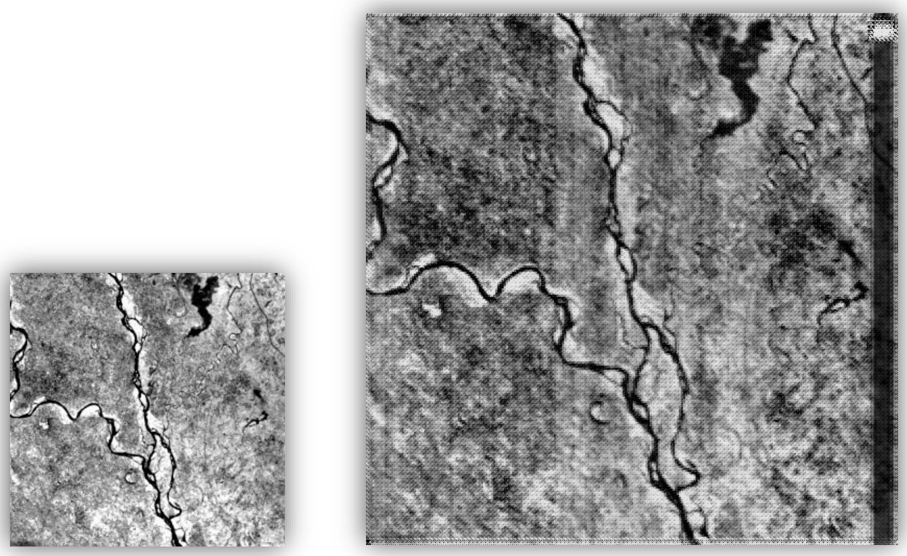
Model Evaluation Results

10.1 Honest Assessment of Results

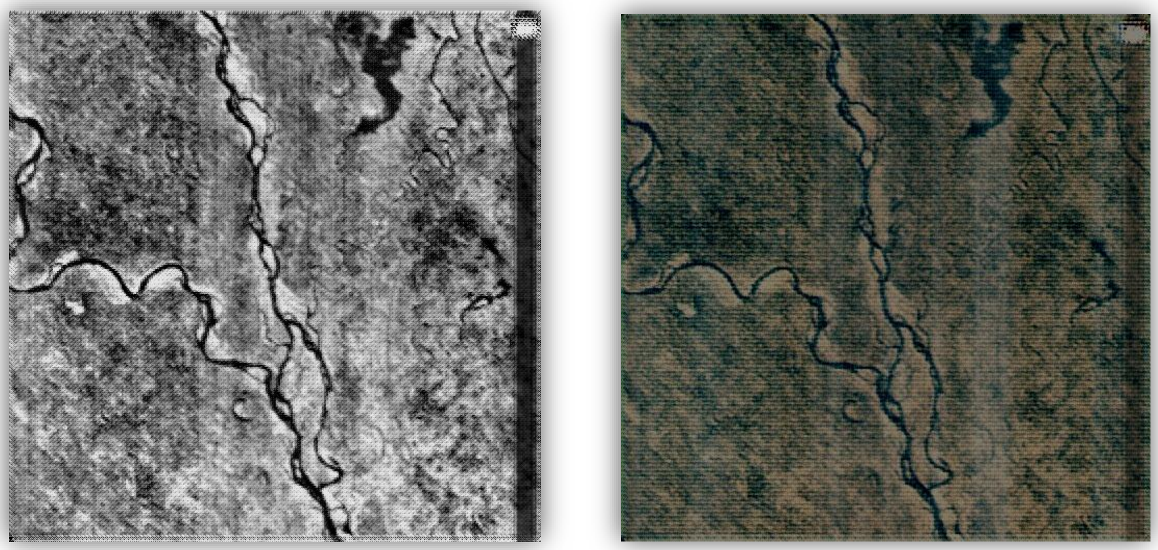
The PSNR and SSIM figures above should be read carefully rather than presented as unqualified success metrics. A PSNR of 18.56 dB and an SSIM of 0.508 are on the low-to-moderate end for image reconstruction tasks; values in this range typically indicate that the generated output captures broad structure but diverges meaningfully from ground truth at the pixel and local-patch level. This is not unexpected for the underlying task colorizing a single thermal channel into three plausible RGB channels is fundamentally ill-posed, since a given temperature value does not map to a unique color in the real world but it should be stated plainly rather than implied to be a strong result. Any presentation of this work should frame these numbers as a functioning first pipeline with room for improvement, not as a solved problem.

10.2 Sample Result Chain

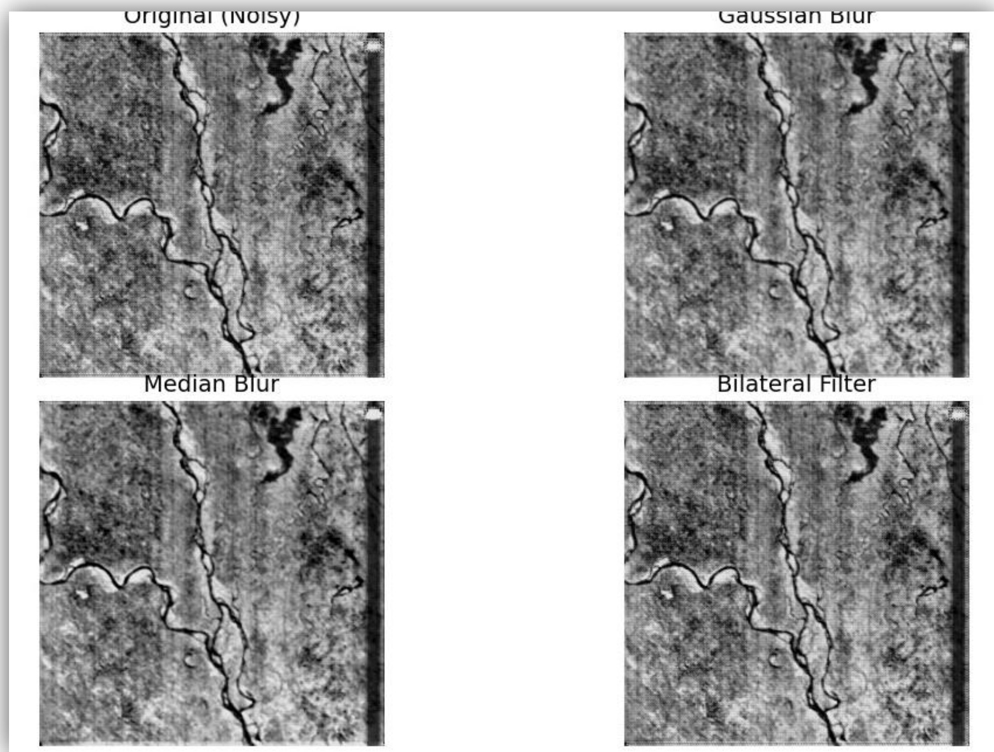
```
Raw Thermal Image         → result/tir.npy

                          → result/tir.png


Super-Resolved Thermal    → result/SR.npy

                          → result/SR.png


Colorized RGB Image       → result/RGB.npy

                          → result/RGB.png
```



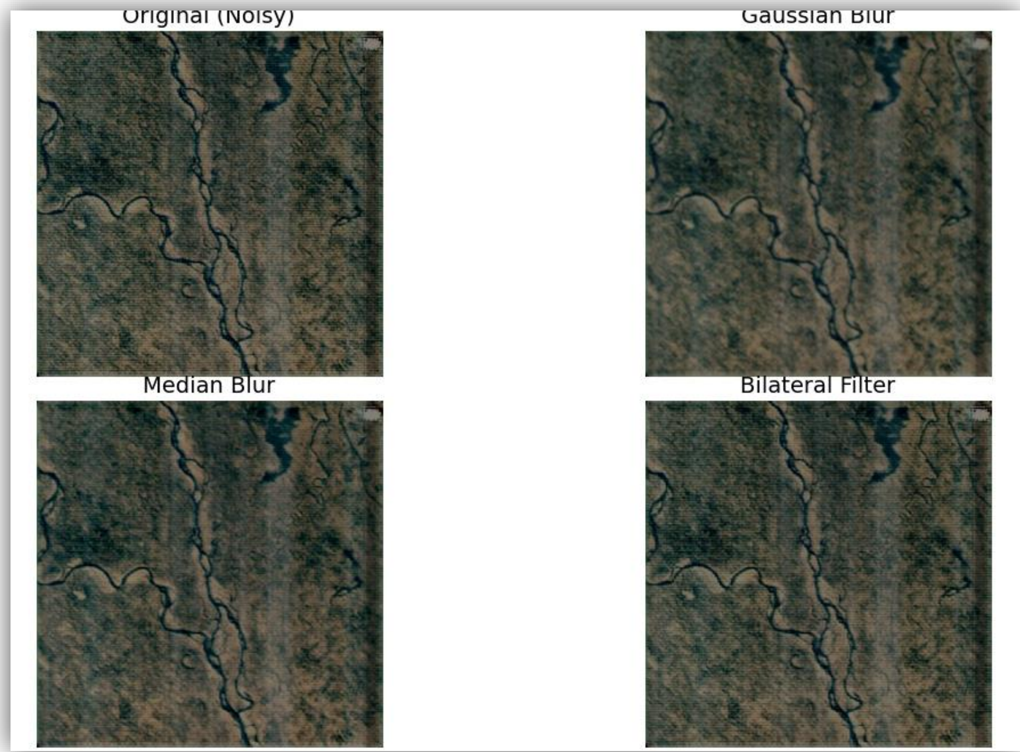
TIR low resolution (1 x 256 x 256) to TIR super resolution (1 x 512 x 512)



TIR super resolution (1 x 512 x 512) to RGB colorized (3 x 512 x 512) -> (512 x 512 x 3)



We can add various blurry method to remove extra Noice



We can add various blurry method to remove extra Noice

11. Limitations and Known Issues

- Limited geographic diversity: training and testing span only three locations (Delhi, Jammu & Lucknow), all within a similar climatic and geographic band generalization to other terrain types (coastal, high-altitude, dense urban megacities) is untested.
- Small test set: evaluation metrics are computed over 16 images from a single test location, which limits confidence in how representative the reported PSNR/SSIM values are.
- Architecture labelling inconsistency between the Stage 1 breakdown and the end-to-end pipeline diagram needs to be resolved in the source documentation.

12. Key Features

- End-to-end deep learning pipeline from raw TIR input to colorized RGB output.
- Independent, modular super-resolution and colorization stages.
- Pix2Pix conditional GAN architecture across both stages.
- Physics-informed feature injection (normalization, local thermal diffusion, gradient magnitude) ahead of colorization.
- Radiometric information preserved via NumPy-based data storage rather than lossy image formats.
- Modular, extensible codebase structure.

13. Applications

- Wildfire detection and tactical fire-perimeter tracking through smoke cover.
- Disaster impact assessment, including night-time flood and land-water boundary mapping.
- Urban heat island analysis and building material heat-retention profiling.
- Environmental and agricultural monitoring.
- Defence, surveillance, and night-time Earth observation.
- General remote sensing research applications requiring interpretable thermal data.

14. Future Work

- SwinIR-based transformer super-resolution to replace or augment the current U-Net generator.
- Transformer-based colorization architectures.
- Attention mechanisms within the generator to improve fine-detail reconstruction.
- Additional physics-aware loss terms tied to material-specific thermal behaviour.
- Multi-spectral feature fusion beyond the current B2/B3/B4/B10 band set.
- Self-supervised pretraining on larger unlabelled thermal datasets.
- Diffusion-based colorization as an alternative to the GAN approach.

16. Conclusion

This project demonstrates a working, physics-informed GAN pipeline that connects raw thermal satellite data to an interpretable, colorized RGB representation, addressing both the spatial resolution and interpretability limitations of native TIR imagery. The dual-stage design separating super-resolution from colorization keeps each network focused on a single, well-defined transformation, and the physics feature injection in Stage 2 gives the colorization network explicit structural cues rather than relying on raw intensity alone.

The current evaluation metrics indicate a functioning but not yet high-fidelity pipeline, and the documentation inconsistencies flagged in Sections 8, 9, and 12 should be resolved before this material is presented to reviewers. With that groundwork addressed, the architecture provides a reasonable foundation for the future improvements outlined in Section 15.

17. Team

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